

SHANTANU VICHARE

■ LEAD QA ENGINEER, AI QUALITY ENGINEERING

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PROFESSIONAL SUMMARY

Lead QA engineer with ~11 years across SaaS quality, test automation, and applied AI. Own quality strategy for PowerSchool's Schoology cloud platform (zero PO incidents over 4+ years) and ship production AI agent systems (GitHub Copilot Background Agents orchestrating Playwright, Browser, and Jira MCPs) that cut manual QA effort by 80%. Champion self-healing automation, cloud-native testing, and DevSecOps (OWASP ZAP, Burp Suite, SAST, DAST) in CI/CD. MSc AI and ML in progress, Liverpool John Moores University.

CORE COMPETENCIES

AI Quality Engineering	Agent-Based Test Automation	Self-Healing Test Frameworks	Cloud-Native Testing Strategies
DevSecOps in CI/CD	Test Strategy and Mentorship	Playwright / WebdriverIO / API Testing	
Quality Observability and Dashboards			

WORK EXPERIENCE

POWERSCHOOL

■ 2020 - PRESENT
Bengaluru, India

Lead Quality Assurance Engineer

- **Scope expansion:** Promoted into AI-enabled Quality Engineering leadership in 2023 after 3 years leading Core QA; now own AI test tooling strategy for Schoology while mentoring 20+ engineers across 6 Agile squads.
- **Agentic QA ecosystem:** Architected a multi-agent test platform with Playwright, Browser, and Jira MCPs orchestrated by GitHub Copilot Background Agents and LLMs (Claude, OpenAI, Gemini) for autonomous test authoring, execution, and ticket triage across Schoology.
- **Self-healing AI tests:** Engineered an AI Context Graph anchoring LLM agents to live OpenAPI contracts and canonical suites, so Playwright tests auto-repair drifted locators on UI change with 90%+ first-pass accuracy and zero hallucinated steps.
- **Autonomous API test synthesis:** Designed an agent-driven pipeline that generates, validates, and runs Playwright API tests from OpenAPI specs, retiring manual Postman workflows and lifting regression coverage from 55% to 95%.
- **80% QA effort reduction:** Scaled Cursor- and Copilot-driven agent workflows across the tribe, compressing prompts-per-task from 15 to 2-3 and reclaiming 4+ engineer-weeks per release cycle.
- **Cloud-native test strategy:** Orchestrated containerized Playwright + WebdriverIO runners across ephemeral cloud environments with parallel workers, cutting regression runtime by 70% and unlocking same-day production deployments.
- **DevSecOps in CI/CD:** Embedded OWASP ZAP DAST, Burp Suite checks, dependency audits, and policy gates into Bamboo, GitLab, and GitHub Actions pipelines, blocking vulnerable code at merge time.
- **Zero-incident SaaS quality (4+ years):** Directed QA strategy for Schoology, sustaining zero PO production incidents while delivering 20+ complex epics on schedule.
- **Quality observability cockpit:** Shipped a MERN-stack internal release tool unifying CI, defect, and test-coverage telemetry through live dashboards, compressing release checklist time from ~4 hours to ~15 minutes.

FIREEYE (MANDIANT)

2018 - 2020
Bengaluru, India

Software Development Engineer in Test

- **Security product test frameworks:** Developed UI and security testing frameworks from scratch for Linux, Windows, and Mac endpoint security products, cutting release escape defects ~35% release-over-release.
- **E2E automation at scale:** Composed Python + Nosetests suites validating detection and prevention across 8+ security releases, shortening regression cycles ~40%.
- **Cross-functional quality:** Partnered with developers and product managers across 5+ distributed teams to triage and resolve complex security software bugs.

- **Methodology uplift:** Spread Python, Selenium, and Robot Framework practices, raising the baseline quality bar for 15+ incoming engineers.

QUICK HEAL TECHNOLOGIES

Associate Threat Research Engineer

2015 - 2018

Pune, India

- **Automation-first malware triage:** Engineered internal tooling that replaced manual malware analysis workflows, reclaiming ~12 analyst-hours per week for higher-value threat research.
- **Incident response and RCA:** Led customer escalation and root-cause analysis across 50+ security incidents, feeding remediation insights back into the product detection roadmap.
- **Product security features:** Proposed 2 threat-detection capabilities shipped into Quick Heal products, strengthening endpoint protection for millions of users.

SELECTED PROJECTS

- AI CONTEXT GRAPH ARCHITECTURE**
Developed an internal grounding framework anchoring LLM test generators to live API contracts and canonical suites: the reusable backbone of the self-healing QA platform at PowerSchool, adopted by 3+ product teams.
- SPECIALIST QA AGENT SUITE**
Built specialist downstream agents for accessibility (WCAG 2.1), internationalization (i18n), and security NFR validation plugged into the multi-agent QA platform, automating compliance checks that previously required dedicated manual specialist review.
- DEVSECOPS PIPELINE BLUEPRINT**
Delivered a reusable CI/CD security blueprint combining OWASP ZAP, Burp Suite, SAST, and dependency audits as gated stages in Bamboo, GitLab, and GitHub Actions, catching 30+ vulnerabilities pre-production.

EDUCATION

- MSc, Machine Learning and Artificial Intelligence – Liverpool John Moores University, England2025 - 2027
- Post Graduate Diploma, AI and ML – IIIT Bangalore2025 - 2026
- Bachelor of Engineering, Information Technology – Shah and Anchor Kutchhi Engineering College, Mumbai2012 - 2015

CERTIFICATIONS

- CopilotXcelerate: AI-Powered Python for Data Science – upGrad2023
- Certified Ethical Hacker (CEH) – Simplilearn2019
- Oracle Certified Expert, Database SQL – Oracle2016

SKILLS

- AI and Agent Tooling:** Cursor, Claude Code, Playwright MCP, Browser MCP, Jira MCP, GitHub Copilot Background Agents, VS Code Copilot, RAG, Multi-Agent Orchestration, LLMs (Claude / OpenAI / Gemini), AI Context Graphs, Self-Healing Test Generation
- Test Automation:** Playwright, Selenium, WebdriverIO, Robot Framework, PHP Codeception, Nosetests, BDD, API Testing, Visual Regression Testing
- Cloud and CI/CD:** Bamboo, GitLab, GitHub Actions, Containerized Test Runners, Parallel Workers, Ephemeral Environments, Cloud-Native SaaS Testing
- Security and Performance:** OWASP ZAP, Burp Suite, OWASP Top 10, DevSecOps, SAST, DAST, Dependency Scanning, JMeter, LoadRunner, Penetration Testing
- Engineering:** Python, TypeScript, JavaScript, PHP, MERN Stack
- Process:** Agile and Scrum, Test Strategy and Planning, Mentorship, Risk Assessment, Vendor Management, Compliance
- Languages:** English, Hindi, Marathi